

Reviewer Pack — Minimal Local Defect Complexity of Affine Quotients vs AC0 P...

Entry 724e91166955 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Defect Complexity of Affine Quotients vs AC0 Parity Circuit Size

Entry ID: 724e91166955 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-24 07:58:09 UTC

1. Conjecture

Field A (mathematical branch): Affine Geometry (Affine Quotients) **Field B** (complexity object): Complexity Theory: AC0 Parity Complexity

Statement:

[‘For any AC0 parity circuit C with size s and n inputs, there exists a parameterization of affine quotients Q such that the minimal local defect complexity of Q is at least $\alpha(s) \cdot \log(n)$, where α is a function independent of n .’, ‘The local defect complexity of an affine quotient Q is defined as the minimum number of elements in a set S such that the closure of the subring generated by S in the field of fractions of Q has transcendence degree at least $\log(|S|)$.’, ‘For any AC0 parity circuit C , it can be shown that there exists an affine quotient Q with minimal local defect complexity satisfying the above condition.’]

Rationale (proposer’s reasoning):

[‘Affine geometry provides a rich structure to study algebraic properties of circuits. The local defect complexity captures the complexity of the algebraic structure of an affine quotient, which is closely related to the size and complexity of AC0 parity circuits.’, ‘If this conjecture holds, it would provide a new invariant that can be used to analyze the complexity of AC0 parity circuits, potentially leading to better lower bounds than those based on random restrictions.’, ‘The bridge between affine geometry and complexity theory could expose new structures and techniques for proving circuit complexity lower bounds.’]

Taxonomy category: AC0_PARITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): d53822b3f09f0015

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The ratio of the minimal local defect complexity to $\log(n)$ is at least $\alpha(s)$, where α is a function independent of n .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "affine geometry" AND "AC0 parity complexity" - "minimal local defect complexity" AND "affine quotients" - "transcendence degree" AND "AC0 circuit size"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_ac0_parity_circuit(n, s):
        # Generate a random AC0 parity circuit with n inputs and size s
        circuit = []
        for _ in range(s):
            gate_type = random.choice(['AND', 'OR'])
            if gate_type == 'AND':
                inputs = random.sample(range(n), 2)
            else:
                inputs = random.sample(range(n), 2)
            circuit.append((gate_type, inputs))
        return circuit

    def compute_local_defect_complexity(circuit):
        # Compute the local defect complexity of the affine quotient Q
        n = len(circuit)
        S = set()
        for gate in circuit:
            if gate[0] == 'AND':
                S.add(gate[1][0])
                S.add(gate[1][1])
            else:
                S.add(gate[1][0])
                S.add(gate[1][1])
        return len(S)
```

```

def alpha(s):
    # Define the function  $\alpha(s)$ 
    return 2 * s

n = random.randint(5, 40)
s = random.randint(5, 40)
circuit = generate_ac0_parity_circuit(n, s)
local_defect_complexity = compute_local_defect_complexity(circuit)

metric_value = local_defect_complexity / math.log(n)
conjecture_holds = metric_value >= alpha(s)
counterexample = "" if conjecture_holds else f"n={n}, s={s}"

return {
    "metric_name": "local_defect_complexity_ratio",
    "metric_value": metric_value,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_metric_value = sum(result["metric_value"] for result in results) / len(results)
    std_metric_value = math.sqrt(sum((result["metric_value"] - mean_metric_value) ** 2 for result
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"n={results[0]['instances_tested']}, s={results[0]['counterexample']}\"")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'counterexample': 'n=21, s=36'
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 6.2275726127014845, 'instances_tested': 1}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 4.061963103757406, 'instances_tested': 1}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 2.5694917118487535, 'instances_tested': 1}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 6.83468428015573, 'instances_tested': 1}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 6.697508766979821, 'instances_tested': 1}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 4.096076519820768, 'instances_tested': 1}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 5.773059132592365, 'instances_tested': 1}

```

```

TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 5.7707801635558535, 'instance': 1}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 6.976382831890591, 'instance': 2}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 6.113218894111955, 'instance': 3}
TRIAL: {'metric_name': 'local_defect_complexity_ratio', 'metric_value': 6.0002541057009395, 'instance': 4}
RESULT: FALSIFIED counterexample="n=1, s=1" first_failing_seed=11

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test has only considered a very small number of instances ($n \leq 15$). The conjecture involves the relationship between AC0 parity circuit size and affine quotient complexity, which may not hold for larger values of n . Without testing larger sizes, it is premature to conclude that the conjecture is false.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test has provided a counterexample where the ratio of the minimal local defect complexity to $\log(n)$ is less than $\alpha(s)$, which meets the falsificati | next: Further investigation is needed to determine if the conjecture holds for larger values of n . Additional testing with larger AC0 parity circuits and affine quotients should be conducted.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12127
2	preregistration	ollama_remote	glm4:latest	0	0	5583
3	novelty	ollama_remote	glm4:latest	0	0	4567
4	novelty	ollama_remote	glm4:latest	0	0	5599
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13077
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6301
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7879
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8693
9	critic	ollama_remote	glm4:latest	0	0	10911
10	judge	ollama_remote	glm4:latest	0	0	6083

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 80820 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/724e91166955.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice

3. The full LLM transcripts are in `research/audit/724e91166955.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/724e91166955.tar.gz` (if generated)